

**Convolutional Neural Network Classification of DaTSCAN Images for Parkinson's Disease
with Interpretability Analysis**

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*I pledge that I have neither given nor received unauthorized aid, including AI, on this
assignment.*

Abstract

Parkinson's disease (PD) is a progressive, neurodegenerative disorder defined by dopamine loss in the striatum, a region of the brain that plays a central role in movement control. Dopamine loss leads to common motor symptoms, such as tremors, rigidity, and bradykinesia. Visualization of dopamine transporter activity through DaTSCAN SPECT imaging has become an increasingly used tool for distinguishing PD from other movement disorders. However, scan interpretation can vary among experts, motivating the development of more consistent automated tools. This study seeks to develop an interpretable dual-input convolutional neural network to classify PD in DaTSCAN images using standardized DaTSCAN images from the Parkinson's Progression Markers Initiative (PPMI). Preprocessed projection images and auto-selected axial-slice images are passed simultaneously through a shared, ImageNet-initialized EfficientNet-B0 feature extractor. The extracted features are combined and passed through a small classification head with Dense layers, Dropout, and a Sigmoid activation function to generate a PD vs. non-PD prediction. Performance is evaluated using receiver operating characteristic area under the curve (ROC-AUC), accuracy, sensitivity, and specificity, all of which are compared against a baseline logistic regression. Interpretability is assessed using Grad-CAM and occlusion testing, which show that the network focuses on biologically relevant features. The model achieved an AUC of 95.5% and an accuracy of 90.7%, demonstrating that interpretable dual-input CNNs can meaningfully enhance consistency in PD imaging assessment and serve as a valuable component to clinical diagnosis.

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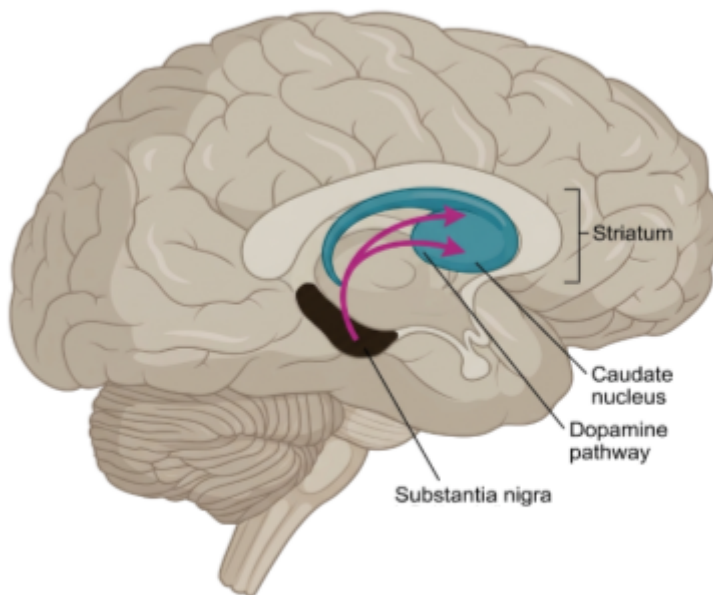
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1. Introduction

Parkinson's disease (PD) is a progressive neurodegenerative disorder in which the neurons that produce dopamine gradually deteriorate and die (Dhivyaa et al., 2024). Dopamine is a chemical messenger in the brain that coordinates smooth and controlled movement (Hathaliya et al., 2022). As neurons die, dopamine levels fall, and the brain loses its ability to regulate movement effectively (Figure 1). Dopamine loss produces the cardinal motor symptoms of PD, including resting tremor, muscle rigidity, and slowed movement (Tolosa et al., 2021). PD is the second most common neurodegenerative disorder in the world, and its global prevalence is projected to double by 2040 as increases in life expectancy bring more people into the age ranges where PD risk is the highest (Kobylecki, 2020; Roussakis et al., 2013). Despite the growth of PD diagnoses, there is still no definitive clinical test for living patients. Diagnosis relies primarily on physicians observing patients' symptoms and medical history—an inherently subjective approach

Figure 1

Illustration of the Nigrostriatal Dopamine Pathway



Note. In Parkinson's disease, degeneration of dopamine neurons in the substantia nigra reduces dopamine signaling to the striatum. Created with BioRender.com.

(Dhivyaa et al., 2024). Several other movement disorders produce symptoms that closely resemble those of PD but do not involve dopamine transporter loss, making clinical diagnosis unreliable on observation alone. These disorders include Essential Tremor, a condition characterized by involuntary shaking, and drug-induced Parkinsonism, which can occur as a side effect of certain

medications (Tolosa et al., 2021). Post-mortem studies of clinically diagnosed PD have shown an overall diagnostic accuracy of 81%, meaning that roughly one in five patients may be misdiagnosed (Rizzo et al., 2016). Early and accurate diagnosis is critical because it enables timely treatment that can slow the progression of PD symptoms and improve quality of life (Dhivyaa et al., 2024).

To support clinical diagnosis and help distinguish PD from similar conditions, physicians increasingly rely on a specialized brain scan called a DaTSCAN, which is a type of single-photon emission computed tomography (SPECT) imaging (Mortezazadeh et al., 2021).

The scan works by injecting a patient with a small amount of a radioactive tracer called Ioflupane I-123, a compound that travels through the bloodstream, crosses into the brain, and binds selectively to dopamine transporter proteins in the striatum (Roussakis et al., 2013). Since the tracer binds to these specific proteins, the resulting image shows the extent of remaining dopamine transporter activity (Roussakis et al., 2013). In a healthy patient, the striatum appears bright and symmetric on the scan, resembling two small comma shapes on either side of the brain. In a patient with PD, the

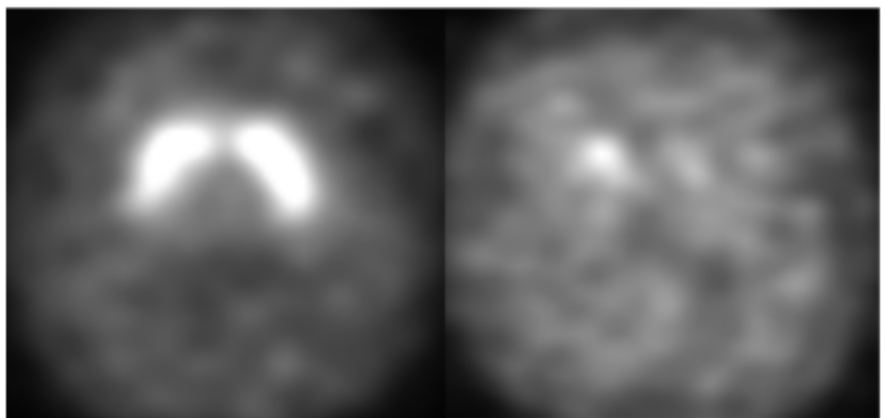
signal is reduced and often asymmetric, reflecting the loss of dopaminergic neurons (Mortezazadeh et al., 2021).

The differences between these scans are shown in Figure 2.

Conditions that have similar symptoms to PD but do not involve dopamine transporter

Figure 2

DaTSCAN Images Comparing Normal and Parkinson's Disease Dopamine Transporter Uptake



Note. Left: healthy patient showing symmetric, comma-shaped striatal uptake. Right: PD patient showing reduced and asymmetric dopamine transporter activity.

loss produce a normal-looking scan, allowing physicians to use DaTSCANS to rule them out when results are abnormal (Roussakis et al., 2013). Despite its advantages over clinical observation alone, DaTSCAN interpretation still requires a physician to visually assess dopamine representation, and this assessment can vary considerably depending on the clinician's experience and expertise (Roussakis et al., 2013). The inconsistency in scan classification limits the reliability of DaTSANs as a standalone diagnostic tool and motivates the development of more consistent, automated approaches to scan analysis.

To develop more objective and consistent approaches to DaTSCAN interpretation, researchers have turned to machine learning, a branch of artificial intelligence that trains computer algorithms to recognize patterns in data by being exposed to large numbers of examples rather than following rules written by a programmer (Dhivyaa et al., 2024). In the same way that a radiologist develops pattern recognition through years of experience, a machine learning model learns by example—but without the subjectivity of human judgment. A specialized subset of machine learning, the Convolutional Neural Network (CNN), has become the dominant approach for analyzing medical images (Wang et al., 2021). The CNN processes an image by passing it through a series of layers that learn to detect increasingly complex visual features (Wang et al., 2021). The learning process starts with visual edges and contrast in the earlier layers and builds toward more impactful structures, such as the shape and symmetry of the striatum (Dhivyaa et al., 2024). A CNN's ability to learn disease-associated scan features directly from image data, without requiring a human expert to manually define measurable criteria, makes it particularly well-suited to medical image analysis (Dhivyaa et al., 2024). In 2017, Choi et al. showed that end-to-end CNNs can interpret DaTSCAN images at a level comparable to that of expert readers (Choi et al., 2017). Building upon this, a 2025 study by Pant et al. discovered that using a pretrained CNN as a feature extractor can achieve strong diagnostic

performance without requiring large datasets or extensive computational resources, making automated classification increasingly practical for real-world applications (Pant et al., 2025).

Despite this progress, the field still faces two important challenges. First, machine learning models applied to DaTSCAN images do not always use a reproducible, standardized preprocessing pipeline, making it difficult to quantify whether performance differences across studies reflect genuine methodological advances or simply differences in how scans were prepared. Second, even when a model achieves high accuracy, it is often unclear which image features drive its predictions and whether those features correspond to the striatal signal loss known to characterize PD or to irrelevant artifacts in the scan, such as brightness or background noise that vary across patient scanners.

This study aims to determine whether a CNN can serve as a reliable and interpretable tool for classifying PD versus non-PD from standardized DaTSCAN images, and to evaluate whether its predictions can be traced to brain regions known to be affected by the disease. The primary research question is: can a convolutional neural network classify PD vs non-PD from standardized, preprocessed PPMI DaTSCAN imaging, and how can its predictions be interpreted and validated using a baseline model and feature-visualization methods? The hypothesis is that a convolutional neural network trained to differentiate between PD and non-PD in DaTSCAN images will establish a unified, systematic approach for using these scans to support a diagnosis.

2. Methods

This project develops a functional diagnostic imaging model that analyzes DaTSCAN SPECT brain scans to differentiate PD from non-PD cases. The model was developed through an iterative testing and improvement process. Success was defined by three criteria: achieving high

classification accuracy, surpassing a logistic regression baseline model, and producing decision patterns driven by anatomically relevant regions rather than incidental image properties.

2.1. Materials

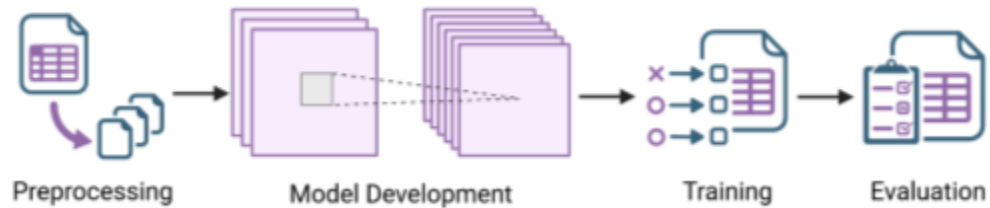
This project uses a combination of neuroimaging datasets, computing hardware, software libraries, and analysis tools to develop and evaluate the CNN model. The data source being used is the Parkinson's Progression Markers Initiative (PPMI) DaTSCAN SPECT dataset, which contains 3,800-4,000 three-dimensional Digital Imaging and Communications in Medicine (DICOM) neuroimaging patient images. These scans include Parkinson's disease cases, Scans Without Evidence of Dopaminergic Deficit (SWEDD) cases—for patients with PD symptoms but no dopamine degeneration, and healthy control cases. The scans are also paired with clinical diagnosis labels and demographic metadata. Model development and training were conducted on a Windows laptop equipped with a 12th Gen Intel Core i7-1260P processor, an Intel Iris Xe Graphics GPU, and 32GB of RAM. Preprocessing, model training, and performance evaluation are performed in Python 3.10 using Visual Studio Code. TensorFlow 2.15 is the primary deep learning framework, supported by NumPy, Pandas, and Scikit-learn for data handling and mathematical calculations. OpenCV and NiBabel are used to load DICOM volumes and extract axial slices. Matplotlib and Google Sheets will be used for visualizing results, including performance curves and interpretable outputs. The code is stored and version-controlled through a [GitHub repository](#).

2.2. Overview of Model Development

This model was developed through four sequential stages: preprocessing, model design, training, and evaluation. Each stage is described in detail in the following sections (Figure 3).

Figure 3

Overview of Model Development Stages



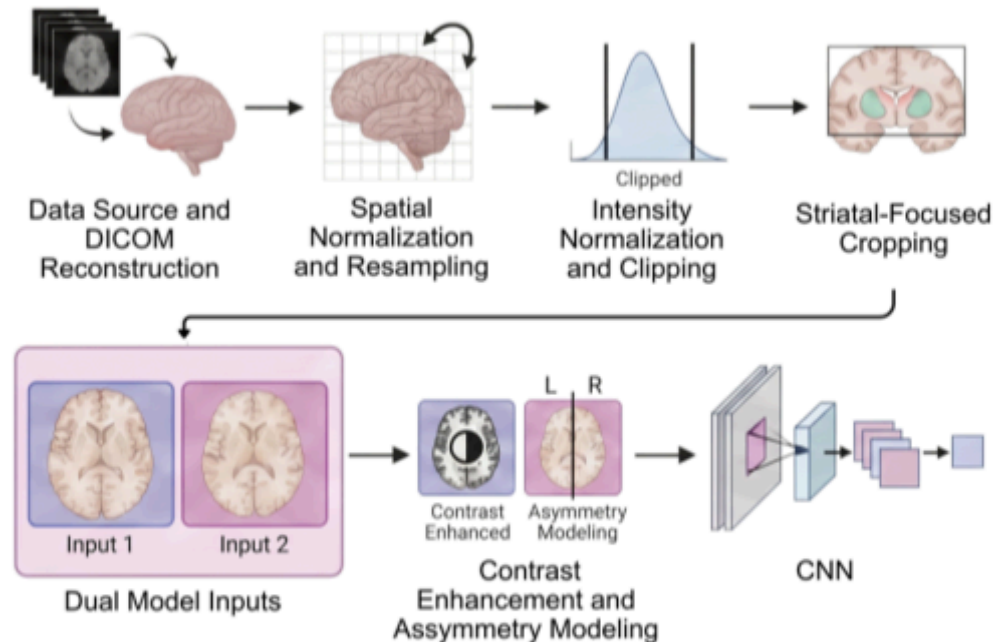
Note. The four sequential stages of model development: preprocessing, model design, training, and evaluation. Created with Biorender.com

2.3. Preprocessing

Each scan underwent a standardized sequence of preparation steps before being passed to the model (Figure 4).

Figure 4

Preprocessing Pipeline



Note. Raw DaTSCAN DICOM images were resampled to uniform voxel spacing, intensity-normalized using percentile clipping, and cropped to a centered striatal region of interest. Created with Biorender.com

2.3.1 Data Source and DICOM Reconstruction

The dataset consists of 960 DaTSCAN brain images collected through the PPMI dataset, comprising 480 PD and 480 non-PD cases. Each scan was paired with a binary label indicating the patient's diagnosis, derived from the metadata file that linked each patient ID to their clinical data classification. Each patient's imaging data was stored as a series of DICOM slices, each representing a thin cross-section of the brain at different depths. These slices were read and stacked in order from bottom to top to reconstruct the full three-dimensional brain volume for each patient. In cases where all slices were stored together in a single file, that file was unpacked directly into the same three-dimensional structure. Otherwise, individual slices were sorted by their depth position and stacked into a three-dimensional array.

2.3.2 Spatial Normalization and Resampling

DaTSCAN images were acquired across different scanners and protocols, so the same anatomical structures appeared at slightly different sizes or resolutions depending on the equipment used. To ensure consistency across patient scans, each three-dimensional volume was resampled to a fixed isotropic voxel spacing of 2.0mm x 2.0 mm x 2.0 mm. A voxel is the three-dimensional equivalent of a pixel, representing a small cube of space within the brain image. Resampling was performed using linear interpolation, a method that smoothly estimates new voxel values when the image grid is resized, avoiding the blocky distortions that would result from simply copying or rounding values. This step ensured that brain structures such as the striatum were spatially aligned across all scans, which is essential for consistent region-of-interest selection and for training a model that expects uniform input dimensions.

2.3.3 Intensity Normalization and Clipping

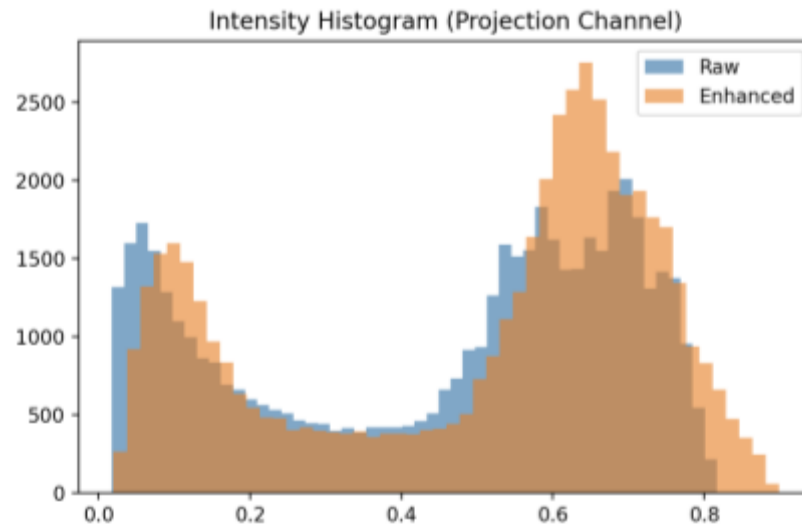
Each scan volume was normalized to prevent unusually bright or dark voxels from dominating the model. First, voxel intensities were percentile-clipped: values below the 2nd percentile were raised, and values above the 99th percentile were lowered. This clipping step removed extreme outliers while still preserving overall uptake patterns in the striatum. After clipping, intensities were scaled to the range [0,1] using min-max normalization (Figure 5). This scaling process standardized the dynamic range across subjects, ensuring the model received consistent input scales across all patients, stabilizing training and improving comparability. The normalization formula is defined as:

$$I' = \frac{\min(\max(I, P_2), P_{99}) - P_2}{P_{99} - P_2 + \epsilon}$$

Where I is the original voxel intensity, P_2 and P_{99} are the 2nd and 99th percentile intensities of the volume, I' is the normalized intensity, and ϵ is the small constant added to avoid division by zero when the clipped range is extremely narrow. The numerator shifts the clipped values so that the lowest allowed intensity becomes zero, and the denominator scales the range so that the highest allowed intensity becomes one. As a result, the full intensity range within each scan is standardized, while the relative spatial contrast is preserved for model training.

Figure 5

Projection Channel Intensity Histogram



Note. Intensity distributions before and after preprocessing for the projection channel, showing the effect of percentile clipping and normalization on scan contrast.

2.3.4 Striatal-Focused Cropping

To ensure that the model focused on an anatomically relevant signal, each scan was cropped to the region centered on the striatum. Focusing the input on this region reduced background noise and discouraged the model from learning patterns driven by irrelevant peripheral features or visual noise rather than disease-related signals.

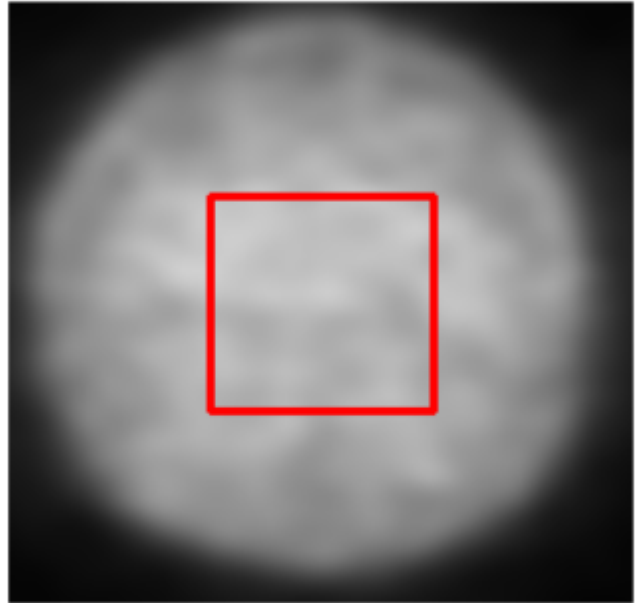
To estimate the center of striatal uptake, the three-dimensional volume was collapsed along the depth axis to produce a single flat image summarizing signal intensity across the brain slices. The produced summary image aggregated signals across slices, stabilizing the striatum's localization even when individual slices were noisy. The image was then thresholded to retain only the top 10% of intensities, isolating the brightest voxels, which are most likely to correspond to striatal dopamine binding. From these high-uptake coordinates, the median x and y

positions were computed. The median, rather than the average, was used to reduce sensitivity to outliers and provide a stable estimate of the striatal center (Figure 6).

Once the striatal center was identified, a fixed 120 x 120 pixel crop was taken around that point, ensuring that every subject had the same spatial dimensions and covered both the left and right striata while maintaining some surrounding context. In the depth direction, a fixed slice range spanning from positions 12-48 was extracted. This range was selected to capture the axial slices with the strongest striatal uptake in the DICOM image, while excluding the superior or inferior regions where little PD-relevant signal is present. The result was a standardized three-dimensional region that focuses the model's attention on biologically relevant signals while maintaining consistent input dimensions across subjects.

Figure 6

Example of Centered Striatal Region of Interest

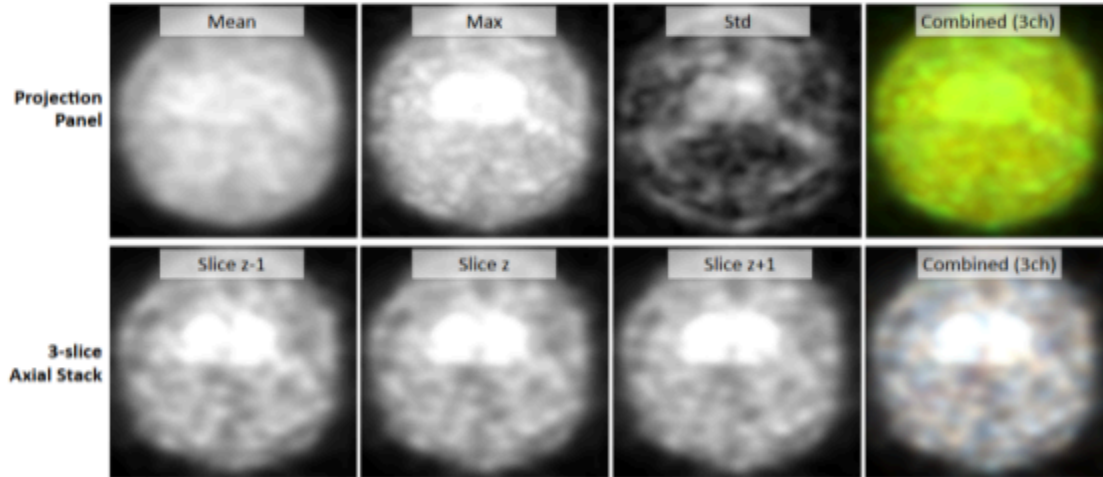


Note. The red box indicates the 120 x 120 pixel crop window centered on the estimated striatal location, used to focus the model on the biologically relevant region.

2.3.5 Dual Model Inputs

Figure 7

Preprocessed Model Inputs for a Single Subject



Note. Final standardized inputs after intensity normalization, resizing, and input construction. Top row shows the projection view; bottom row shows the three-slice axial stack.

Each subject was represented by two complementary inputs, enabling the model to learn both a global summary of striatal signals and a spatially localized view of the brain (Figure 7). This dual-input design maximized the biological information available to the model while remaining compatible with standard two-dimensional CNN architectures. In the first input, the full three-dimensional scan volume was compressed into a single two-dimensional projection image, which captured the overall distribution of dopamine transporter uptake across slices. Three different statistical projections were computed and stacked as separate channels: mean, max, and an asymmetry map. The mean projection captured average uptake levels across the brain, the max projection highlighted peak signaling in the striatum, and the left-right asymmetry map computed the absolute difference between the mean projection and its mirror image. The asymmetry channel compares the pixel intensity values to highlight hemispheric imbalance, an early imaging sign of PD. The second input was a stack of three slices centered around the

highest dopamine uptake. Keeping the slices as separate channels preserved information that would otherwise be lost if the projections were collapsed into one image. This three-channel format also matched the expected input dimensions of the model's feature extractor, EfficientNet-B0.

For the projection input, let $V(z,x,y)$ be the cropped three-dimensional volume, where z indexes axial slices and (x,y) is the in-plane location. The projection channels were computed by collapsing the volume across the axial dimension:

$$\mu(x,y) = \frac{1}{Z} \sum_z V(z,x,y), \quad M(x,y) = \max_z V(z,x,y), \quad \sigma(x,y) = \sqrt{\frac{1}{Z} \sum_z (V(z,x,y) - \mu)^2}$$

Here, Z is the number of axial slices in the cropped volume. The mean projection $\mu(x,y)$ averaged uptake across all slices at each (x,y) location, reducing noise and improving signal estimation. The max projection $M(x,y)$ captured the strongest uptake observed at each location, preserving the focal high-intensity regions rather than diluting them through averaging. The standard deviation projection $\sigma(x,y)$ quantifies the signal's variation across slices, helping the model distinguish regions of consistent uptake from those that fluctuate with depth. These summaries were stacked to form the projection input, providing the model with a compact but informative view of the three-dimensional volume.

For the slice-stack input, the axial slice with the highest mean intensity was selected using the following function:

$$k = \arg \max_z \left(\frac{1}{HW} \sum_{x,y} V(z,x,y) \right),$$

Where H and W are the in-plane height and width, this function identifies the slice most representative of peak striatal dopamine uptake. The model then took the center slice and its two neighbors and stacked them as separate channels:

$$X_{slices}(x, y) = [V_{k-1}(x, y), V_k(x, y), V_{k+1}(x, y)]$$

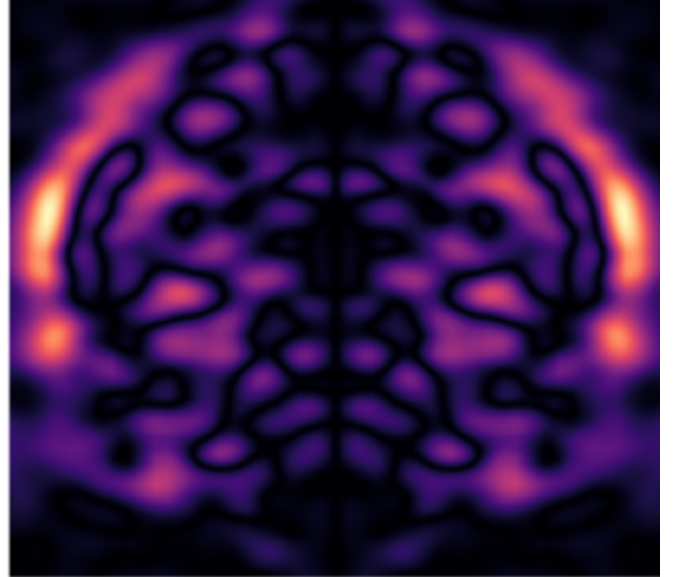
This approach, known as a 2.5D representation, preserved the striatal signal across adjacent slices, allowing the model to capture depth information without the computational cost of a 3D CNN. Both the projection input and slice stack input were resized to 244 x 244 pixels before entering the network to meet the input dimensions required by EfficientNet-B0.

2.3.6 Contrast Enhancement and Assymetry Modeling

To strengthen the striatal signal and test augmentation-enhanced inputs, a secondary preprocessing stage applied Contrast Limited Adaptive Histogram Equalization (CLAHE) and unsharp masking to each channel. This technique improved local contrast while limiting noise amplification in homogeneous regions, particularly in low-contrast nuclear imaging, thereby adding clarity to striatal boundaries without altering symmetry. Unsharp masking added a controlled edge enhancement by subtracting a lightly blurred version of the image and adding the difference back, further emphasizing striatal edges and central uptake patterns.

In addition to contrast enhancement, an asymmetry channel was also computed to capture left-right differences, which were clinically relevant because early PD often showed asymmetric dopaminergic loss. After the mean projection was computed, each pixel was paired with its mirror across the image's midline. If the image width is W , then the

Figure 8
Example Asymmetry Channel Image



Note. Computed as the absolute difference between each pixel and its horizontal mirror across the brain's midline. Brighter regions indicate greater left-right imbalance in dopamine transporter activity, which is an early imaging sign of PD.

mirror of (x, y) is $(x, W - 1 - y)$, where μ is the intensity value of the image at a specific pixel.

The asymmetry value at (x,y) is the absolute difference between the two mirrored intensities

(Figure 8):

$$A(x, y) = |\mu(x, y) - \mu(x, W - 1 - y)|$$

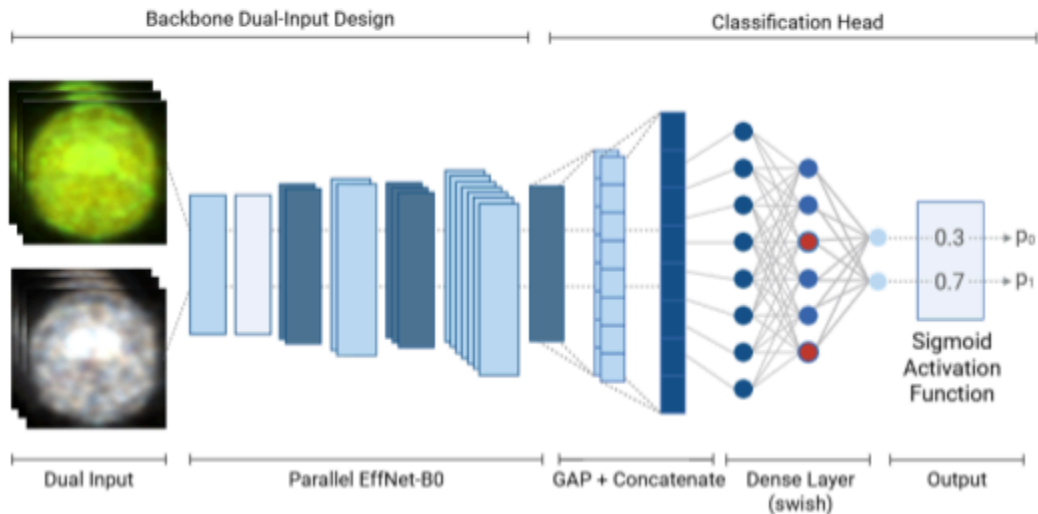
This formula produces a heatmap in which bright regions indicate the strongest hemispheric imbalance. In PD, uptake loss is often asymmetric, so this channel allowed the model to directly observe that imbalance rather than inferring it from raw intensity patterns. The asymmetry map was then stacked with the mean and max projections as the third channel.

2.4. Model Development

Model design was centered around balancing classification accuracy with computational accessibility and interpretability (Figure 9).

Figure 9

Dual-input CNN Architecture

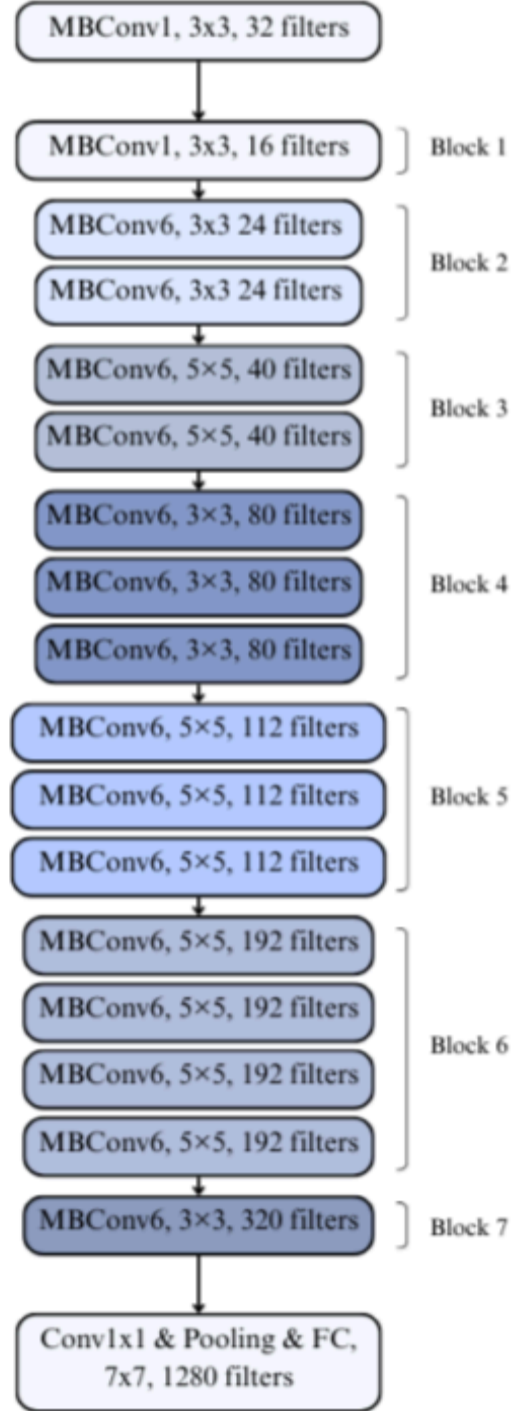


Note. Both scan inputs are processed simultaneously through a shared EfficientNet-B0 backbone. The resulting feature vectors are concatenated and passed through a classification head to produce a PD probability score. Created with BioRender.

2.4.1 Backbone Dual-Input Design

The model was built using a shared EfficientNet-B0 backbone, a type of CNN architecture that processes images through a series of sequential blocks. EfficientNet-B0 was chosen due to its balance of accuracy and efficiency. The projection image and a slice-stack image were passed simultaneously through the same EfficientNet-B0 feature extractor. The architecture was initialized with weights pre-trained on ImageNet, a large database of general images, which provided a starting point for recognizing basic visual patterns such as edges, contours, and textures. The model's layers were then fine-tuned on the DaTSCAN data, with the filters adjusted to focus on patterns distinguishing PD scans from non-PD scans. EfficientNet-B0 consists of nine main blocks for processing (Figure 10). Seven of these blocks contain Mobile Inverted Bottleneck Convolution (MBConv) layers, which expand channel dimensions, apply depthwise convolution, and then project the features back to a compact representation (Alhichri et al., 2021). This design enabled the network to capture complex visual

Figure 10
EfficientNet-B0 Architecture



Note. Nine sequential processing blocks, seven of which use Mobile Inverted Bottleneck Convolution layers. Early blocks detect simple visual features such as edges and contrast; later blocks identify more complex structures.

patterns more efficiently, reducing computational costs. The filter dimensions describe the size of the sliding filter that detects certain features in the image (Figure 10). Smaller filters capture finer and more detailed structures, while larger filters detect broader anatomical relationships and spacing within the image. The number of filters (e.g., 15, 24, 40) indicates the number of distinct feature types the model learned at each stage. The early EfficientNet-B0 blocks detect simple elements such as edges or intensity variations, while the latter blocks identify more specialized structures, including striatal symmetry, uptake intensity gradients, and abnormal dopaminergic patterns associated with PD.

If x_p is the projection input and x_s is the slice-stack input, the corresponding feature vectors are defined as:

$$z_p = f_0(255x_p), \quad z_s = f_0(255x_s), \quad z_p, z_s \in \mathbb{R}^{1280}$$

Here, $f_0(\cdot)$ denotes EfficientNet-B0 as a feature extractor with global average pooling, so its output is a single 1280-dimensional vector. The inputs are multiplied by 255 because they are stored in the $[0,1]$ range, while EfficientNet-B0 is calibrated for images in the 0-255 range. The two feature vectors are concatenated as $z = [z_p; z_s] \in \mathbb{R}^{2560}$ before entering the dense classification head. Each input branch was fed into EfficientNet-B0, which produced a single feature vector summarizing the network's learned representation of that input. One vector came from the projection image and the other from the slice-stack image.

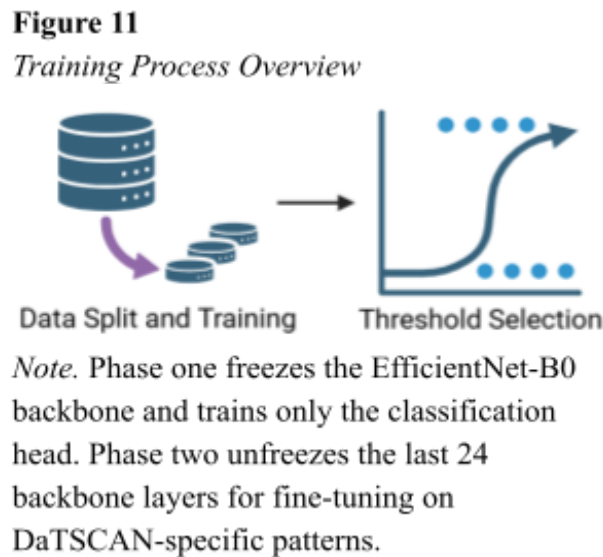
2.4.2 Classification Head

The two feature vectors were concatenated into a single feature vector so the model could examine global uptake patterns and local striatal structure simultaneously. This combined vector was then passed into a fully connected layer with 384 units and a Swish activation function, mixing information from both inputs and allowing the model to learn interactions between the

two feature sets. The layer size of 384 units was chosen because it was large enough to learn meaningful combinations without adding an excessive number of parameters, reducing overfitting on a limited medical dataset. Next, a dropout layer with a rate of 0.4 randomly disabled 40% of the units during training, forcing the model to spread information across multiple neurons rather than relying too heavily on a few, thereby improving generalization to new patients and reducing overfitting. The sigmoid activation function converted the output to a probability between 0 and 1, with the higher values indicating a higher likelihood of PD. This probability was then compared with the chosen decision threshold to produce the final binary PD/non-PD label.

2.5. Training

Model training was conducted in two sequential phases, designed to preserve the pre-trained features of EfficientNet-B0 while gradually adapting them to DaTSCAN-specific patterns (Figure 11).



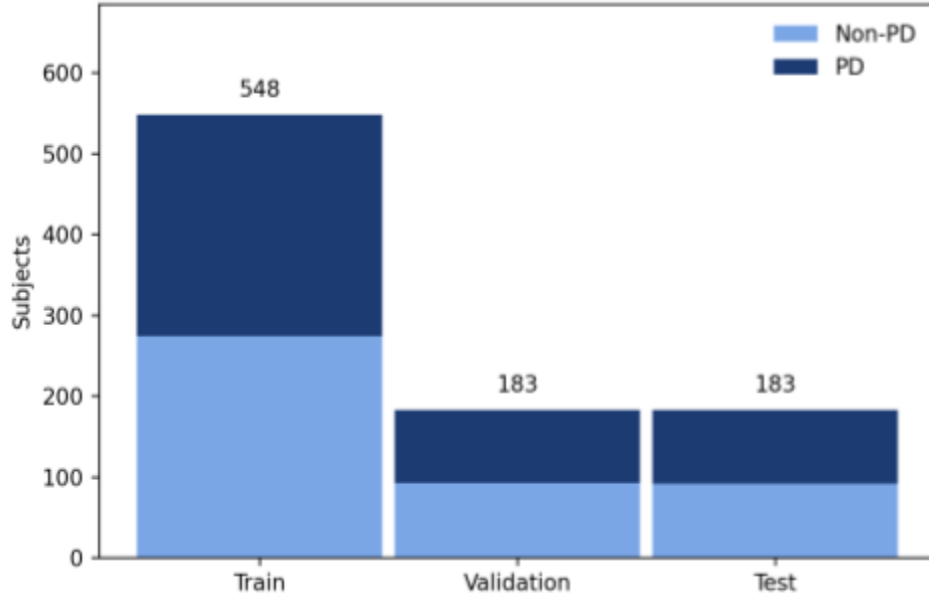
2.5.1 Data Split and Training

The dataset was divided into training, validation, and test sets using a stratified shuffle split. Stratified splitting preserved the class ratio, so the proportions of PD and non-PD cases remained nearly the same across subsets. The shuffle randomly mixed the data before splitting, reducing ordering bias. This splitting strategy ensured that the model is treated as a representative sample during training and that performance estimates on validation and test data were not skewed by class imbalance.

The test set was fixed at 20% of the data. The remaining 80% was then split again, with 25% assigned to validation, resulting in a final split of 60% for training, 20% for validation, and 20% for testing (Figure 12). The validation set was used only for model selection and tuning, while the test set was held out until the end to provide an unbiased estimate of final performance. The final test set contained 183 scans, comprising 92 PD and 91 non-PD cases, resulting in a near-balanced distribution. This balance stabilized estimates of sensitivity and specificity and prevented the larger class from inflating accuracy.

Figure 12

Stratified Training, Validation, and Test Split



Note. Dataset split by subject to prevent data leakage, with class balance preserved across all three subsets: 60% training, 20% validation, 20% testing.

Training was performed in two phases. In phase one, the EfficientNet-B0 backbone was frozen, and only the classification head was trained. This isolation allowed the new layers to learn task-specific decision boundaries without immediately changing the pre-trained features. The head was optimized using Adaptive Moment Estimation (Adam) with a learning rate of 8×10^{-4} and a batch size of 16. Early stopping monitored validation, AUC, and restored the best weights to prevent overfitting. In phase two, the last 24 layers of the backbone were unfrozen for fine-tuning. This step allowed higher-level feature filters to adapt to DaTSCAN-specific signal patterns, but the learning rate was reduced to 2×10^{-5} and the batch size to 12 to keep updates small and controlled. Reduce-on-Plateau automatically lowered the learning rate when validation AUC stopped improving, helping the model converge smoothly.

2.5.2 Threshold Selection

The sigmoid output produces a probability rather than a direct probability, so a decision threshold τ is needed to convert the probability $\hat{p}$ into a binary PD or non-PD prediction. Instead of fixing $\tau = 0.5$, the threshold was selected using the validation set to find the value that maximized model performance. The threshold selection is written as:

$$\hat{y} = 1[\hat{p} \geq \tau], \quad \tau^* = \arg \max_{\tau} \frac{1}{N} \sum_{i=1}^N 1[\hat{y}_i = y_i]$$

The first expression says that the predicted class $\hat{y}$ is 1 (PD) if the predicted probability $\hat{p}$ is at least the threshold τ , and otherwise it is 0. The second expression defines the best threshold τ^* as the one that maximizes validation accuracy. For each candidate threshold, accuracy was computed as the fraction of correctly classified validation examples, i.e., the average of the indicator $1[\hat{y}_i = y_i]$ over all N validation cases. The threshold yielding the highest accuracy was selected and fixed for the final test evaluation, keeping the test set untouched throughout and ensuring a fair evaluation. For transparency aid and comparison to common practice, performance at $\tau = 0.5$ was also reported as a reference.

2.6. Evaluation Metrics

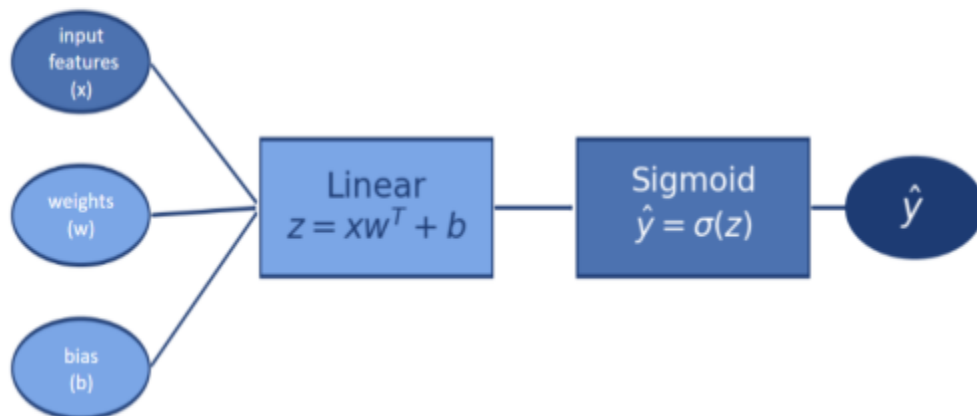
Model performance was measured using four metrics: accuracy, sensitivity, specificity, and ROC-AUC. These metrics were selected together because accuracy alone can mask class-specific errors, making sensitivity and specificity necessary to assess how well the model performs on PD and non-PD cases separately. Table 1 defines each metric. A confusion matrix was also produced to show the raw counts underlying these metrics.

Table 1*Performance Metrics Used to Evaluate Model Classification*

Metric	What It Measures
Accuracy	Overall proportion of scans correctly classified as PD or non-PD
Sensitivity	Proportion of true PD cases correctly identified, reflecting the model's ability to detect disease
Specificity	Proportion of non-PD cases correctly identified, reflecting the model's ability to rule out disease
ROC-AUC	Overall ability to distinguish PD from non-PD across all possible decision thresholds, where 1.0 is perfect, and 0.5 is random chance. A higher AUC indicates that PD and non-PD probabilities are well separated, even if the operating threshold changes.

Note. Sensitivity and specificity were reported alongside accuracy to capture class-specific performance, as accuracy alone can obscure errors in individual groups.

2.6.1 Baseline Comparison

Figure 13*Logistic Regression Baseline Flowchart*

Note. Global intensity statistics extracted from both inputs are combined into a feature vector, mapped to a linear score, and converted to a PD probability using a sigmoid function.

To verify that the CNN adds predictive value beyond simple global intensity statistics, a logistic regression baseline is trained on manually extracted global features from the projection images (Figure 13). For each subject, the mean, standard deviation, minimum, and maximum intensity values are computed across the spatial axis and concatenated into a feature vector $\phi(x)$. This feature vector captures overall signal level and spread, but does not encode spatial structure or left-right partners. The baseline prediction is defined as:

$$\hat{p}_{base} = \sigma(w^T \phi(x) + b)$$

Here, $\phi(x)$ is the feature vector for a scan, w is the learned weight vector, b is the bias term, and $\sigma(\cdot)$ is the sigmoid function that maps the linear score into a probability between 0 and 1. The baseline model essentially makes its decision using a simple weighted combination of global intensity statistics, without seeing the details of the striatum's spatial layout. This baseline serves as a transparent, low-capacity benchmark. If the CNN significantly outperforms this model, it indicates that the CNN is learning meaningful spatial patterns rather than just responding to global brightness or contrast differences.

2.6.2 Interpretability Analysis

Interpretability is evaluated using Gradient-weighted Class Activation Mapping (Grad-CAM), which is a visualization technique that produces a heatmap highlighting the important regions in an image that the CNN used to make a specific prediction. The heatmap is produced based on gradients flowing into a chosen convolutional layer. When applied to DaTSCAN brain images, Grad-CAM is designed to explain model predictions by highlighting the brain regions that most influence the class score. The produced are overlaid on the raw projection channels and slice stacks, allowing for visual confirmation that high-saliency regions correspond to the striatum. This interpretability step is vital because dopamine transporter signal loss in

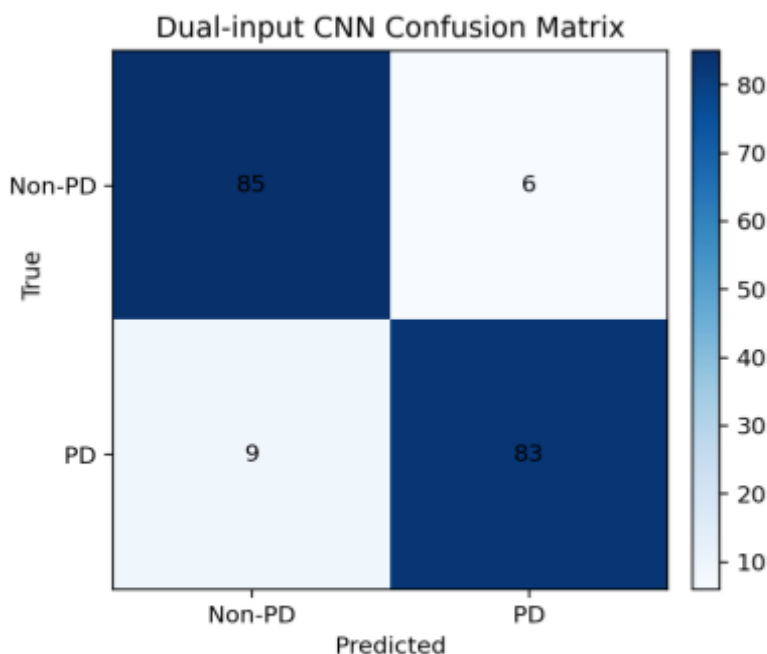
DATSCANs is known to be concentrated in the striatal regions in PD, and a reliable model should emphasize these areas rather than peripheral artifacts. To complement Grad-CAM, occlusion sensitivity testing is used as a second, more direct analysis, wherein small sections of the scan slice are symmetrically occluded, the prediction is recomputed, and the drop in PD probability is recorded. Regions that cause the largest decrease in model confidence when occluded are considered more important, and alignment between these regions and Grad-CAM heatmaps strengthens confidence that the model is using biologically meaningful uptake patterns.

3. Results

The dual-input CNN achieved strong classification performance on the test set. Using the validation-selected operating threshold of 70.5%, the model reached an AUC of 95.5%, an accuracy of 90.7%, a sensitivity of 90.2%, and a specificity of 91.2% (Figure 14).

Figure 14

Confusion Matrix for Dual-input CNN

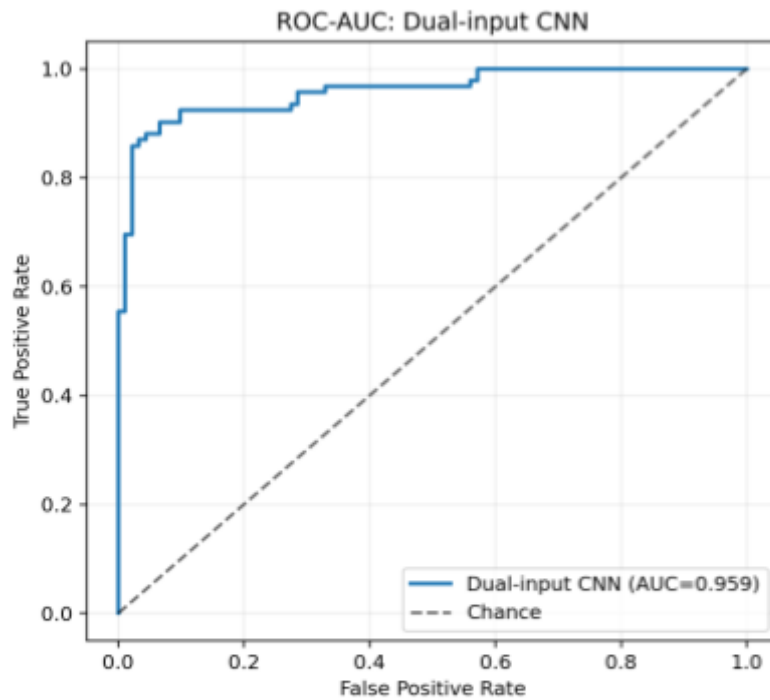


Note. Counts of true positives, true negatives, false positives, and false negatives on the held-out test set, showing a balanced error profile across PD and non-PD cases.

The produced confusion matrix indicates a balanced error profile, with low false-positive and false-negative rates, showing that the model does not favor PD detection at the expense of control classification and maintains reliable performance in both cases (Figure 15).

Figure 15

ROC-AUC Curve for Dual-input CNN



Note. True positive rate plotted against false positive rate across all decision thresholds. The area under the curve was 95.5%, indicating strong separation between PD and non-PD probability scores.

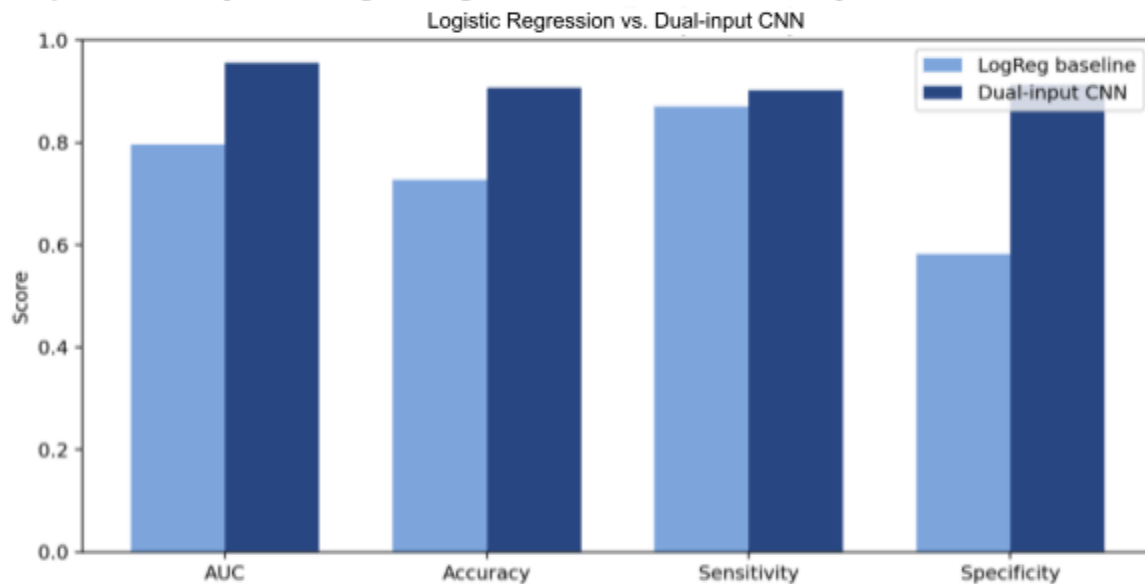
3.1 Comparison to Baseline

The logistic regression baseline model trained on global summary features provides a reference for how much value the CNN adds. The baseline achieved an AUC of 79.6%, an accuracy of 72.7%, a sensitivity of 87%, and a specificity of 58.2%. Relative to this baseline, the dual-input CNN improves AUC by 15.9%, accuracy by 18%, sensitivity by 3.2%, and specificity by 33% (Figure 16). The largest improvement occurs in specificity, demonstrating that the CNN

substantially reduces false positives. Balanced accuracy provides a concise summary of this improvement: 0.726 for the baseline versus 0.907 for the CNN, indicating that the CNN improves overall separability while also correcting the baseline's bias towards predicting.

Figure 16

Performance Comparison: Logistic Regression Baseline versus Dual-input CNN



Note. Accuracy, AUC, sensitivity, and specificity for both models on the held-out test set, showing the CNN's improvement over the baseline across all four metrics.

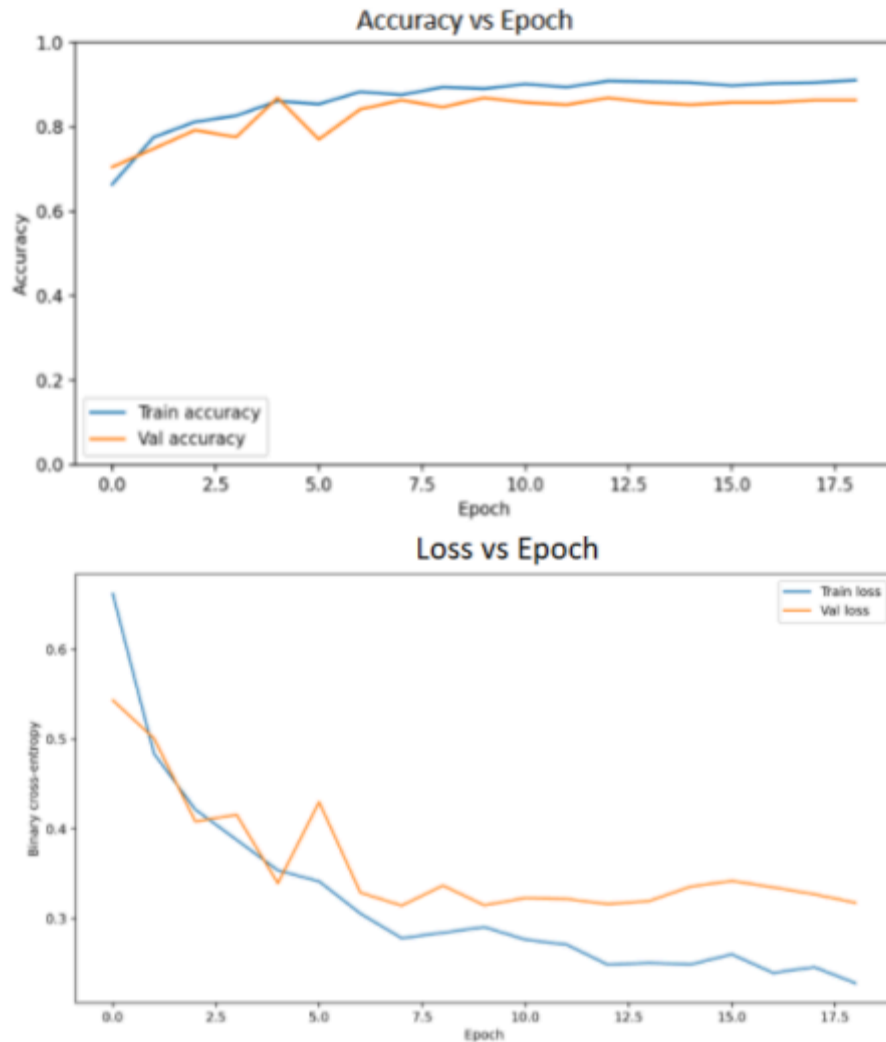
3.2 Thresholding and Training Generalization

The model's probability outputs are informative across thresholds, enabling the operating point to be tuned to clinical priorities. The selected threshold of 0.705 yields the reported accuracy of 0.907, while using a fixed 0.5 threshold reduces accuracy to 0.842.

Training and validation curves show stable convergence and no evidence of overfitting, suggesting that the model generalizes well (Figure 17). The preservation of high validation performance across epochs aligns with the strong test set results and supports the robustness of the learned features.

Figure 17

Training Curves for the Dual-input CNN



Note. Accuracy and loss plotted across training epochs for both the training and validation sets. An epoch is one complete pass of the training data through the model. Stable convergence with

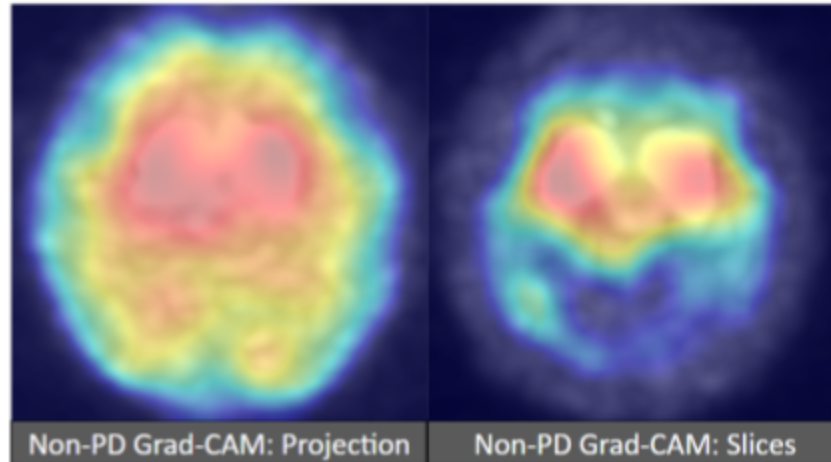
3.3 Interpretability Analysis: Grad-CAM and Occlusion Testing

Grad-CAM analysis reveals that the model's predictions are driven by striatal regions that align with the known dopamine neurodegeneration patterns in PD (Figure 18). In PD-positive cases, the strongest activations are consistently localized to the striatum, reflecting the expected

pattern of signal loss. In non-PD cases, activations are weaker and more symmetric, consistent with preserved uptake.

Figure 18

Grad-CAM Heatmaps for PD and Non-PD Cases

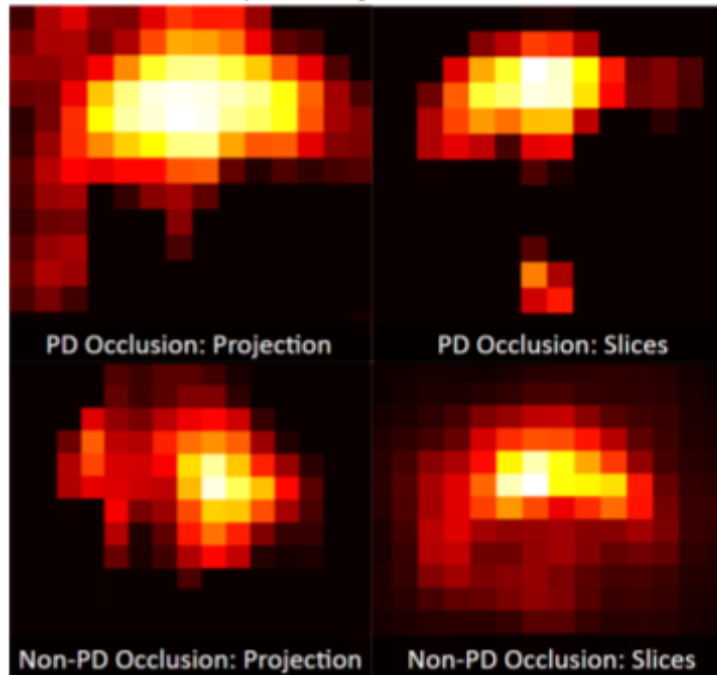


Note. Activation maps overlaid on projection and slice inputs. Warmer colors indicate regions with stronger influence on the model's prediction. PD cases show concentrated activation over the striatum; non-PD cases show weaker and more diffuse activation.

Occlusion sensitivity testing provides a perturbation-based validation of feature importance. The largest decreases in PD probability occurred when the occluded regions overlapped the striatum, indicating that the model's confidence depended most strongly on striatal uptake, with other regions contributing less to the decision (Figure 19).

Figure 19

Occlusion Sensitivity Heatmaps



Note. Each pixel reflects the drop in PD prediction confidence when that region of the scan was blocked. Regions with the largest drops, concentrated over the striatum, were most important to the model's predictions.

4. Discussion

This study developed and evaluated a dual-input CNN to classify PD versus non-PD from standardized DaTSCAN images. The model achieved strong classification performance, supporting the hypothesis that a standardized, interpretable CNN can serve as a reliable tool for automated DaTSCAN classification. The interpretability analysis confirmed that predictions were driven by biologically relevant striatal features instead of incidental image properties, addressing key concerns about the reliability of machine learning tools in clinical settings.

4.1 Overall Classification Performance

The model's AUC of 95.5% and accuracy of 90.7% indicate that a standardized dual-input CNN can serve as a reliable tool for automated DaTSCAN classification, supporting the study's central hypothesis. The balance between sensitivity and specificity, at 90.2% and 92.2%, respectively, makes the model more favorable in a clinical context. A model that heavily favors sensitivity flags too many healthy patients as having PD, while one that favors specificity misses too many true cases. The near-equal error rates suggest that the model performed consistently across both groups. This level of performance was achieved on a standard laptop without cloud computing, demonstrating that strong diagnostic accuracy does not require extensive computational resources.

4.2 Baseline Comparison and Spatial Learning

The logistic regression baseline achieved a specificity of 58.2%, meaning that it misclassified nearly half of all non-PD cases as PD. The CNN increased specificity to 91.2%, a 33% gain. Global intensity statistics alone are insufficient for reliably ruling out PD because they cannot capture the spatial distribution of dopamine transporter activity in the striatum, including its shape, symmetry, and local signal partners. The CNN's large gain in specificity confirms that the dual-input design successfully learned this spatial structure and that processing two complementary image representations simultaneously was justified by the result.

4.3 Threshold Selection and Model Generalization

The validation-selected threshold of 70.5% achieved 90.7% accuracy, compared to 84.2% with the conventional fixed threshold of 0.5. The model's outputs were systematically shifted above 0.5, indicating that a fixed threshold did not align with the model's true probability

distribution. Selecting the threshold on the validation set is correct for this, without using any information from the test set data. The stable training curves and similarities between validation and test set performance indicate that the model learned generalizable features rather than memorizing the training data, an important finding given the relatively small dataset.

4.4 Interpretability and Relevance

Grad-CAM and occlusion testing independently identified the striatum as the primary region driving the model's predictions. Since the two methods work through different mechanisms, one by analyzing internal network gradients and the other by directly measuring the effect of removing image regions, their agreement strengthens confidence that the finding reflects genuine model behavior. The striatum is where dopamine transporter loss is most concentrated in PD, so the model's represented reliance on this region indicates it learned biologically relevant features. A model that achieved high accuracy by responding to scanner artifacts or image formatting differences rather than disease signal would not be dependable in new clinical environments. The interpretability results suggest that the model's decisions were grounded in the same imaging signal that physicians use when reading DaTSCAN images, a vital step towards building clinical trust in automated classification tools.

4.6 Comparison and Broader Context

The results achieved by this model are consistent with the range of performance reported in the DaTSCAN classification literature. Models from Khachnaoui et al. (2023) and Pant et al. (2025) reported higher accuracies with more complex architectures, including multi-backbone designs and ensemble strategies, which require substantially greater computational resources. The model presented here was intentionally designed to remain trainable on a standard laptop,

and a 95.5% AUC under those constraints demonstrates that competitive performance is achievable with a computationally accessible approach. Several prior studies did not report specificity, making a full comparison difficult, but the 92.2% specificity obtained from the proposed model is relevant because a model that cannot reliably rule out PD would generate too many false positives to be clinically useful. Prior work has also shown that machine learning can detect PD from data beyond DaTSCAN imaging. Gait-based classifiers have demonstrated that motor patterns carry disease-relevant information detectable from wearable sensor data (E et al.), and CNN-based structural MRI analysis has shown that PD-related brain changes are visible in modalities that do not directly measure dopamine transporter activity (Reddy). These findings, alongside the results presented here, suggest that PD-related changes are broadly detectable across multiple data types and that standardized, interpretable machine learning tools will play a role across this range of approaches.

Table 2

Comparison of Proposed Dual-Input CNN With Prior PD DaTSCAN/SPECT Classification Studies

Study	Method/Classifier	Dataset/Input	Accuracy	AUC	Sensitivity	Specificity	Notes
This work (2026)	Dual-input CNN (2.5D projection + slice stack; shared backbone)	PPMI SPECT (DaTSCAN)	90.7%	95.50%	90.20%	91.20%	Single model, reduced compute
Wenzel et al. (2019)	Deep CNN	PPMI FP-CIT SPECT	6.70%	NR	NR	NR	3D CNN on full FP-CIT SPECT volumes
Kurmi et al. (2022)	Xception (base learner)	PPMI SPECT (DaTSCAN)	95.3%	NR	98.84%	88.37%	Single CNN base model
Khachnaoui et al. (2023)	BCNN EfficientNet-B0–MobileNet-V2	PPMI SPECT (DaTSCAN)	98.5%	NR	NR	NR	Bilinear CNN, transfer learning
Pant et al. (2025)	SqueezeNet feature extractor + Neural Network classifier	PPMI SPECT (DaTSCAN)	97.3%	99.60%	NR	NR	Transfer learning + ML classifier

Note. NR indicates that the metric was not reported in the original study. All models were evaluated on PPMI DaTSCAN SPECT data. Results for this work were produced from the held-out test set.

4.7 Limitations

A practical limitation of this study is the number of patients available for training and evaluation. The obtained cohort consists of 480 PD and 480 non-PD patients, providing a solid, balanced sample for DaTSCAN work but remaining modest relative to the data demands of deep CNNs. Deep models learn subtle patterns best when they encounter large, diverse populations. With fewer patients, there is less exposure to rare imaging variations, atypical presentations, scanner differences, and demographic diversity. The dataset size does not undermine the validity of the results, but it does mean the performance estimates reflect the limits of the sample. Data augmentation and class balancing help the model generalize, yet they cannot fully replace the variability that comes from larger, multi-site cohorts. Broader datasets would further strengthen the model's confidence in generalization and help determine how well it performs across different clinical environments.

A second limitation is the interpretability of a CNN. This type of neural network model can learn what PD-like striatal uptake patterns look like, but it cannot always specify which visual evidence led to a correct decision. Visualization tools such as Grad-CAM highlight influential regions, and occlusion testing provides a more direct check of importance, yet these methods remain approximations rather than complete explanations. Using visualization tools allows the model to be biologically informative while still being partially opaque about the full decision logic in each case. The lack of definitive visualization tools does not make the model untrustworthy, but it does mean predictions should be interpreted alongside clinical context and further validation, especially for atypical or borderline scans.

A third limitation is computational resources, which directly influenced model design. To keep training feasible, the model used a 2.5D dual-input strategy, projection channels plus slice

stacks, rather than a full three-dimensional volumetric CNN. This tradeoff reduces compute and memory demands but may miss some spatial context that a true three-dimensional model could capture. Limited computation also constrained how extensively hyperparameters, input resolution, and ensemble strategies could be explored.

4.8 Future Work

Future work would prioritize broader, multi-site datasets to test generalization across scanner vendors, acquisition protocols, and patient demographics. Expanding beyond PPMI would help determine how robust the model is in real clinical settings and reduce the risk of dataset-specific performance inflation. A second direction is multimodal PD modeling, combining DaTSCAN with functional biomarkers such as gait or tremor signals. This could improve early detection and enable progression tracking, since functional measures reflect patient impact while DaTSCAN reflects dopaminergic loss. A third extension is to evaluate richer model families and ensembles. With greater computational resources, future studies could explore high-resolution inputs, full three-dimensional CNNs, longer training schedules, and systematic hyperparameter searches. Ensemble approaches such as Fuzzy Rank Level Fusion could also be tested to determine whether diversity across architecture yields additional gains without sacrificing interpretability.

5. Conclusion

This work demonstrates that a dual-input CNN can classify PD versus non-PD with high accuracy from DaTSCAN SPECT images while maintaining biological relevance through interpretability analysis. By combining projection-based summaries with slice-stack inputs, the model captures both global uptake patterns and local striatal structure, consistently

outperforming a transparent baseline based on simple intensity features. Grad-CAM and occlusion sensitivity further show that the model's decisions align with the striatum. These findings contribute to the growing body of evidence that automated, interpretable CNN-based tools have the potential to support more consistent DaTSCAN interpretation and reduce the diagnostic variability that currently affects patient care.

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7. References

- Alhichri, H., Alswayed, A. S., Bazi, Y., Ammour, N., & Alajlan, N. A. (2021). Classification of Remote Sensing Images Using EfficientNet-B3 CNN Model With Attention. *IEEE Access*, 9, 14078–14094. <https://doi.org/10.1109/ACCESS.2021.3051085>
- Choi, H., Ha, S., Im, H. J., Paek, S. H., & Lee, D. S. (2017). Refining diagnosis of Parkinson's disease with deep learning-based interpretation of dopamine transporter imaging. *NeuroImage: Clinical*, 16, 586–594. <https://doi.org/10.1016/j.nicl.2017.09.010>
- Dhivyaa, C. R., Nithya, K., & Anbukkarasi, S. (2024). Enhancing Parkinson's Disease Detection and Diagnosis: A Survey of Integrative Approaches Across Diverse Modalities. *IEEE Access*, 12, 158999–159024. <https://doi.org/10.1109/ACCESS.2024.3487001>
- Hathaliya, J., Parekh, R., Patel, N., Gupta, R., Tanwar, S., Alqahtani, F., Elghatwary, M., Ivanov, O., Raboaca, M. S., & Neagu, B.-C. (2022). Convolutional Neural Network-Based Parkinson Disease Classification Using SPECT Imaging Data. *Mathematics*, 10(15), 2566. <https://doi.org/10.3390/math10152566>
- Khachnaoui, H., Chikhaoui, B., Khelifa, N., & Mabrouk, R. (2023). Enhanced Parkinson's Disease Diagnosis Through Convolutional Neural Network Models Applied to SPECT DaTSCAN Images. *IEEE Access*, 11, 91157–91172. <https://doi.org/10.1109/ACCESS.2023.3308075>
- Kobylecki, C. (2020). Update on the diagnosis and management of Parkinson's disease. *Clinical Medicine*, 20(4), 393–398. <https://doi.org/10.7861/clinmed.2020-0220>
- Kurmi, A., Biswas, S., Sen, S., Sinitca, A., Kaplun, D., & Sarkar, R. (2022). An Ensemble of CNN Models for Parkinson's Disease Detection Using DaTscan Images. *Diagnostics*, 12(5), 1173. <https://doi.org/10.3390/diagnostics12051173>

- Mortezaazadeh, T., Seyedarabi, H., Mahmoudian, B., & Islamian, J. P. (2021). Imaging modalities in differential diagnosis of Parkinson's disease: Opportunities and challenges. *Egyptian Journal of Radiology and Nuclear Medicine*, 52(1), 79.
<https://doi.org/10.1186/s43055-021-00454-9>
- Pant, J., Pant, H. K., Pant, V. K., Bhatt, V., Singh, D., & Joshi, K. (2025). Deep Learning-Based Feature Extraction and Machine Learning Models for Parkinson's Disease Prediction Using DaTscan Image. *Biomedical and Pharmacology Journal*, 18(December Spl Edition), 161–177. <https://doi.org/10.13005/bpj/3079>
- Rizzo, G., Copetti, M., Arcuti, S., Martino, D., Fontana, A., & Logroscino, G. (2016). Accuracy of clinical diagnosis of Parkinson disease. *Neurology*, 86(6), 566–576.
<https://doi.org/10.1212/WNL.0000000000002350>
- Roussakis, A.-A., Piccini, P., & Politis, M. (2013). Clinical utility of DaTscan™ (123I-Ioflupane Injection) in the diagnosis of Parkinsonian Syndromes. *Degenerative Neurological and Neuromuscular Disease*. (world).
<https://www.tandfonline.com/doi/abs/10.2147/DNND.S19807>
- Tolosa, E., Garrido, A., Scholz, S. W., & Poewe, W. (2021). Challenges in the diagnosis of Parkinson's disease. *The Lancet Neurology*, 20(5), 385–397.
[https://doi.org/10.1016/S1474-4422\(21\)00030-2](https://doi.org/10.1016/S1474-4422(21)00030-2)
- Wang, Z. J., Turko, R., Shaikh, O., Park, H., Das, N., Hohman, F., Kahng, M., & Chau, D. H. (2021). CNN Explainer: Learning Convolutional Neural Networks with Interactive Visualization. *IEEE Transactions on Visualization and Computer Graphics*, 27(2), 1396–1406. <https://doi.org/10.1109/TVCG.2020.3030418>
- Wenzel, M., Milletari, F., Krüger, J., Lange, C., Schenk, M., Apostolova, I., Klutmann, S., Ehrenburg, M., & Buchert, R. (2019). Automatic classification of dopamine transporter

SPECT: Deep convolutional neural networks can be trained to be robust with respect to variable image characteristics. *European Journal of Nuclear Medicine and Molecular Imaging*, 46(13), 2800–2811. <https://doi.org/10.1007/s00259-019-04502-5>